

Group Project Week 1:

Structuring a \$1 Million Portfolio: Gold, Silver, and Gold-Mining Equities

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Precious Metals: Many Ways to Invest \$1,000,000



GOLD

- Physical Bullion
- ETFs: GLD, IAU, SGOL
- Funds & Futures



SILVER

- Physical Bullion
- ETFs: SLV, SIL, SIVR
- Funds & Futures



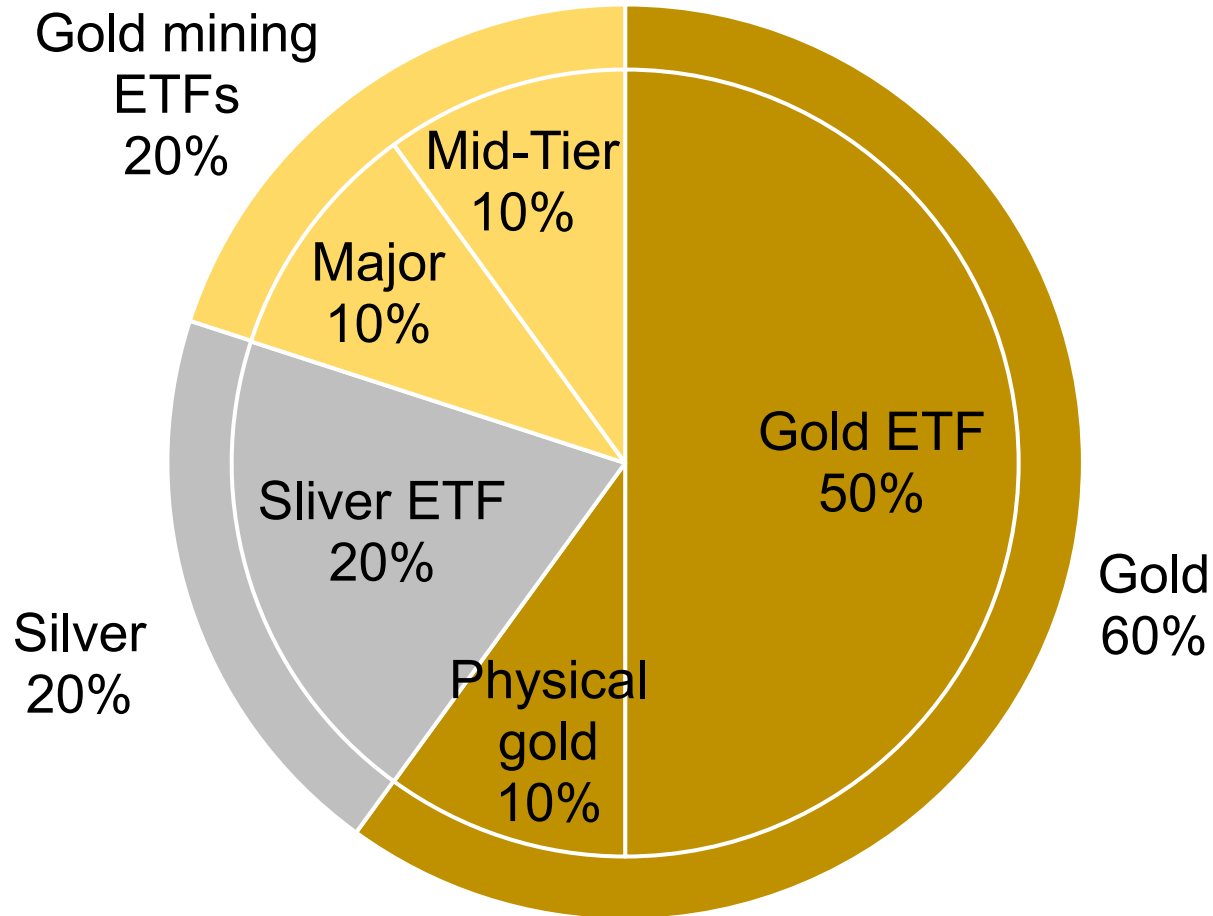
GOLD MINING EQUITIES

- ETFs: GDX, GDXJ, SGDJ
- Majors & Mid-Tiers
- Junior Miners

\$1,000,000 to invest. Countless paths, one recommended mix.

Source: Durrett, D. (2010). How to Invest in Gold and Silver: A Complete Guide with a Focus on Mining Stocks.

Portfolio Solution



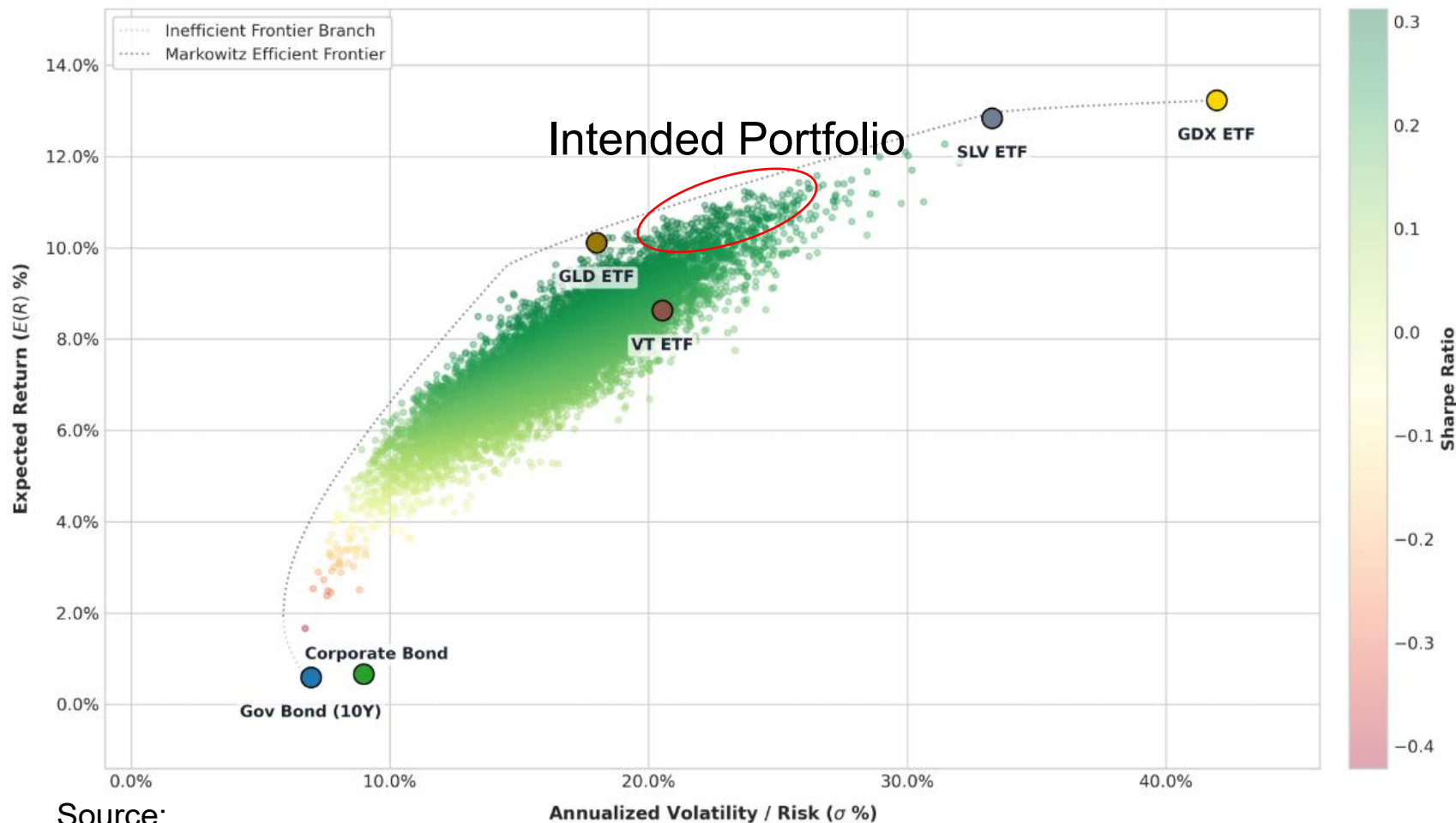
Time horizon: Five years +
Investment philosophy: Long term position

AUM: 1 Million USD

Investment style: Long term

Investment time horizon: 5 years +

Markowitz Portfolio Optimization & Efficient Frontier
2006-08-29 to 2026-08-28



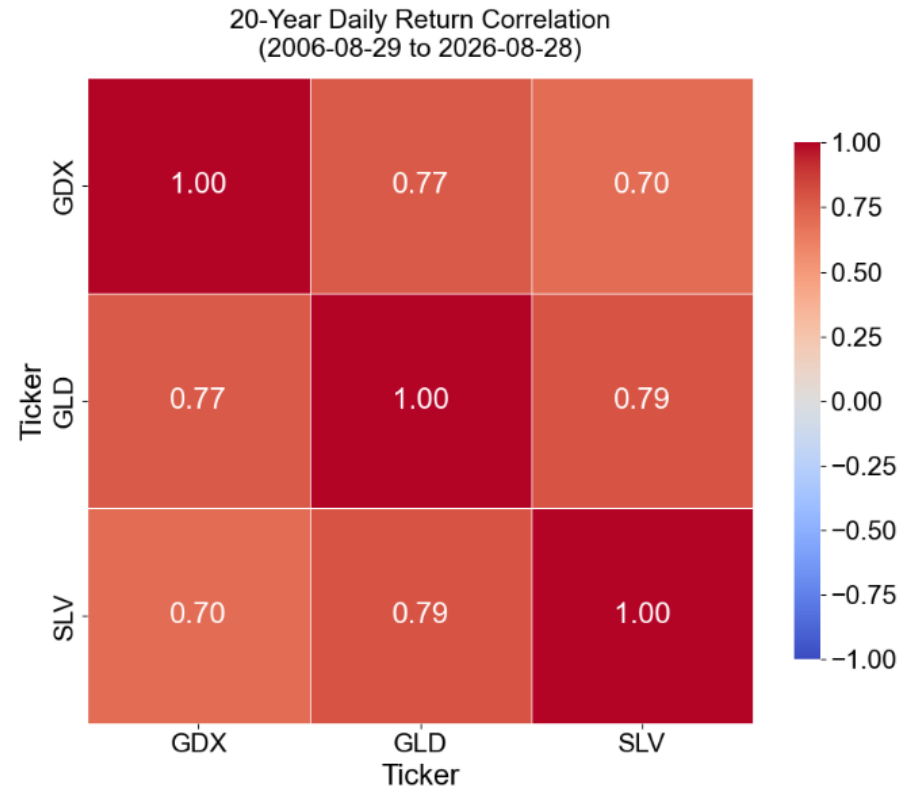
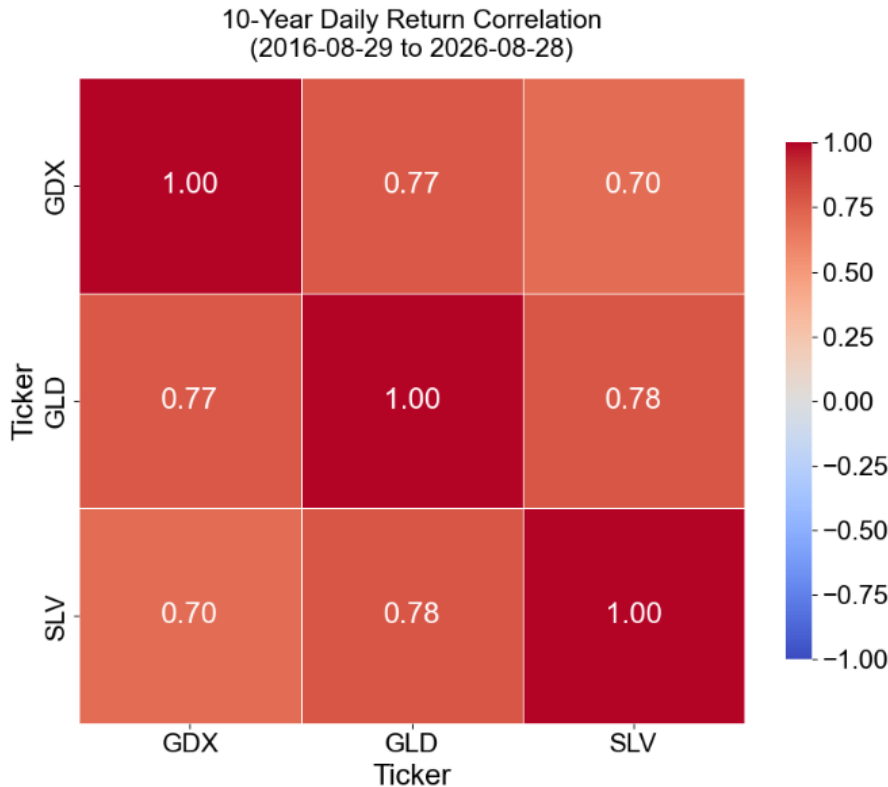
Source:

[1] <https://towardsdatascience.com/python-markowitz-optimization-b5e1623060f5/>

[2] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Portfolio%20CAPM%20simulation.ipynb>

Disclaimer: Historical data is for reference only and does not indicate future performance

Gold vs Silver vs Gold mines are strongly correlated (Using GLD, SLV and GDX as simulation)



Source:

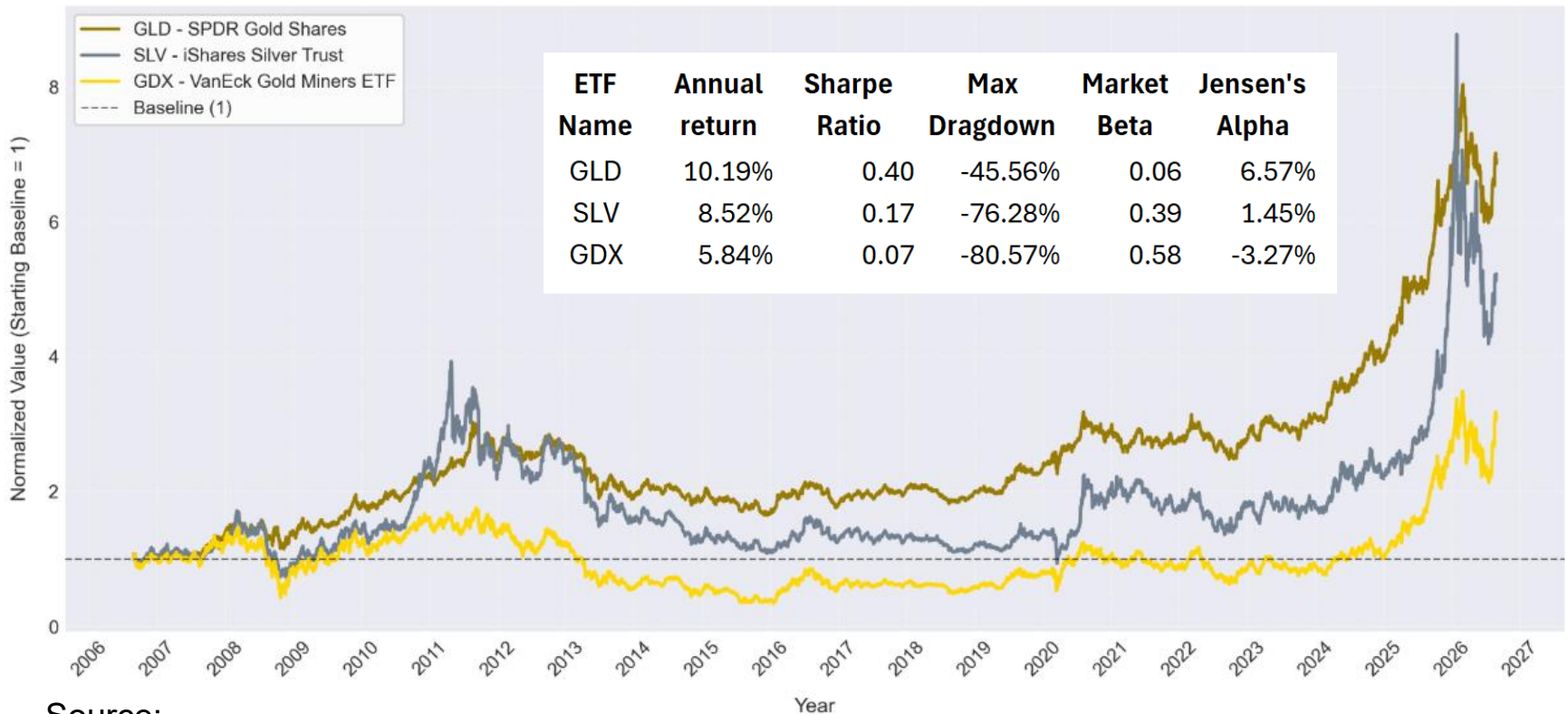
[1] Yahoo Finance: <https://finance.yahoo.com/quote/GLD>; <https://finance.yahoo.com/quote/SLV>; <https://finance.yahoo.com/quote/GDX>

[2] GitHub: <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/FMR%20A%20-%20Correlation%20GLD%20SLV%20GDX.ipynb>

Disclaimer: Historical data is for reference only and does not indicate future performance

Comparison of Gold vs Silver vs Gold mines using historical performance (Using GLD, SLV and GDX as simulation)

20-Year Cumulative Performance Comparison (Base = 100) (2006-08-29 to 2026-08-28)



Source:

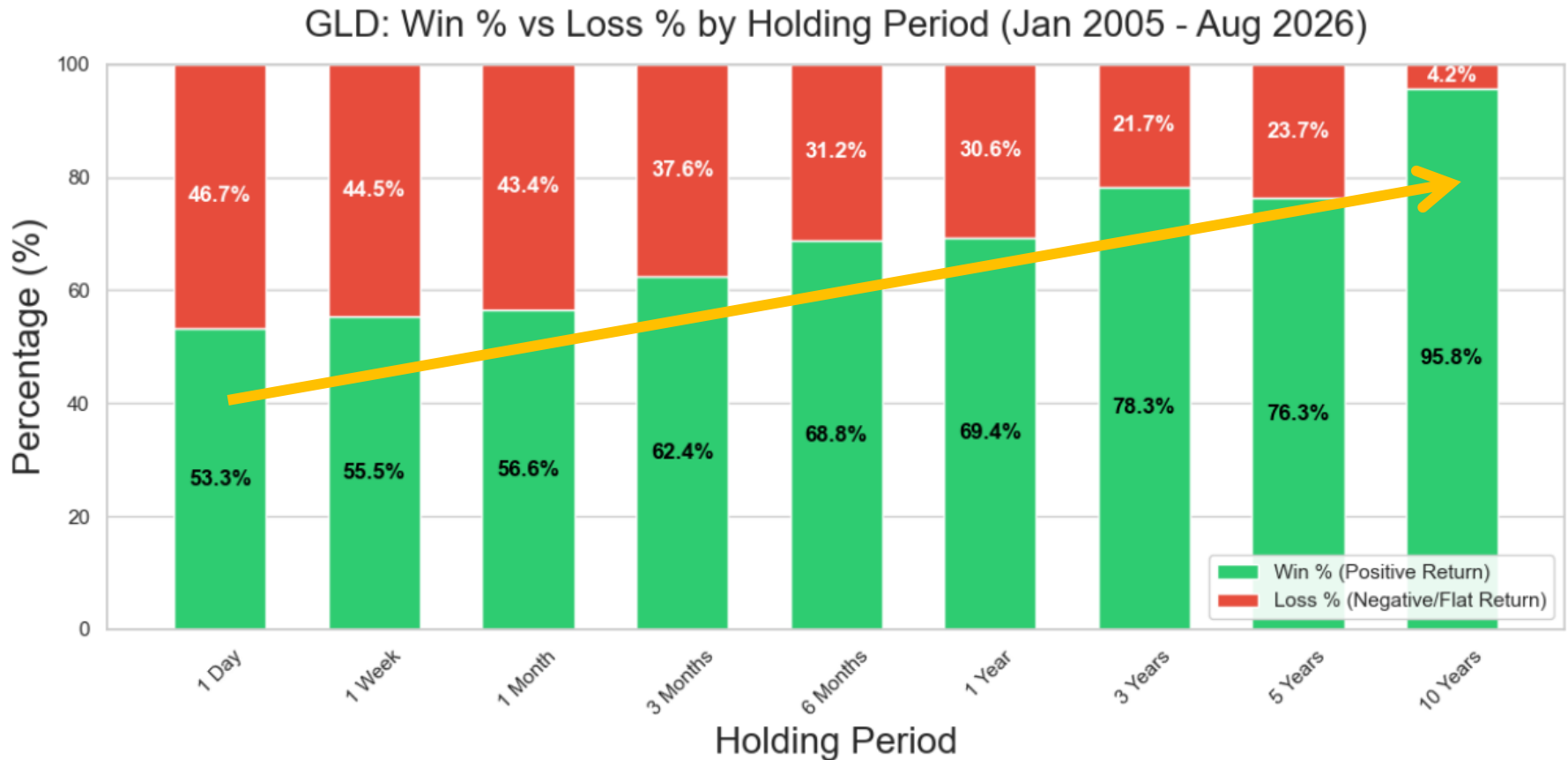
[1] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Potfolio%20ratios.ipynb>

[2] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Historical%20performance%20-%20GLD%20GDX%20GDXJ%20SGDJ.ipynb>

Disclaimer: Historical data is for reference only and does not indicate future performance

Gold as the main asset in the long run

Safe-haven asset, Value preservation, Central bank accumulation, High liquidity and?



The longer you invest, the more likely you are winning

Source:

[1] <https://finance.yahoo.com/quote/GLD/>

[2] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Holding%20GLD%20longer%20corr%20with%20return.ipynb>

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Comparison of gold investment methods

GLD, IAU, GLDM, IAUM, SGOL; GDX, GDXJ, SGDJ; NEM, AEM

Method	Examples	Pros	Cons
Paper Gold (Financial Claims)	Gold ETFs & Funds Mining Stocks Futures & Certificates	High market liquidity Zero storage & vault fees Easy digital trading	Counterparty risk No direct physical claim Market brokerage fees
Physical Gold (Tangible Asset)	Gold bars & Bullion Gold coins Allocated vault storage	Direct asset ownership Zero counterparty risk Inflation safe haven	Storage & insurance cost Theft & security risk Lower immediate liquidity



Investors benefit most from holding mostly paper gold, supplemented by physical gold for immediate liquidity and urgent needs.

Source:

[1] Durrett, D. (2010). How to Invest in Gold and Silver: A Complete Guide with a Focus on Mining Stocks.

[2] <https://cmi-gold-silver.com/inside-the-gold-vault/>

Physical gold choice - Goldmoney.com (Toronto listed company XAU.TO)

“The starting place for investing in precious metals is physical metal. It is recommended to include some physical metal as part of your portfolio.”

– Don Durrett

APMEX.com
JMBullion.com
SDBullion.com
ProvidentMetals.com
BullionExchanges.com
BullionVault.com
Goldmoney.com →

Large distributors + Free insurance & shipping for large orders
Bullion storage required

Recommended reasons:

- Extremely competitive pricing on bulk and wholesale orders
- Best for high volume investors
- Listed company

Source:

[1] Durrett, D. (2010). How to Invest in Gold and Silver: A Complete Guide with a Focus on Mining Stocks.

[2] <https://finance.yahoo.com/quote/XAU.TO/>

Gold ETF choice - GLDM

Gold ETFs: Expense Ratio vs. Avg Daily Dollar Volume



While GLDM is not the cheapest ETF, its high liquidity provides strong value for investors.

Source:

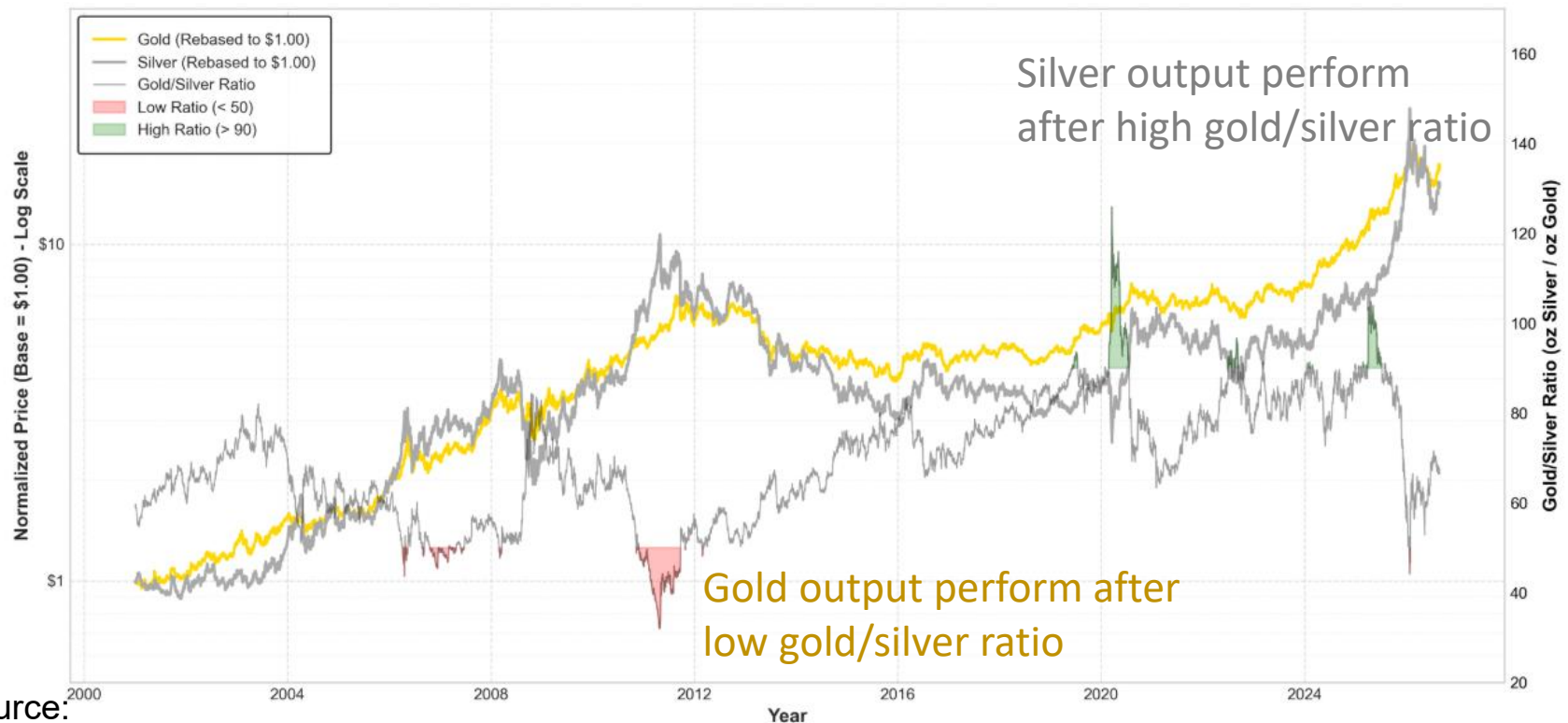
[1] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Gold%20ETF%20comparison.ipynb>

[2] Khomyn, M., Putniņš, T., & Zoican, M. (2024). The Value of ETF Liquidity. *The Review of Financial Studies*, 37, 3092-3148.

Why Silver as minor investment? – Judgement based on Gold Silver ratio

- Investment Demand
 - Safe heaven & Weaker USD
 - Lower real interest rates
- Industrial Demand
 - Electronics & automobile
 - Solar

Historical Performance: Gold, Silver, and Gold/Silver Ratio (Jan 2001 – Aug 2026)



Source:

[1] <https://silverinstitute.org/wp-content/uploads/2026/04/World-Silver-Survey-2026.pdf>

[2] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Gold%20Silver%20Ratio.ipynb>

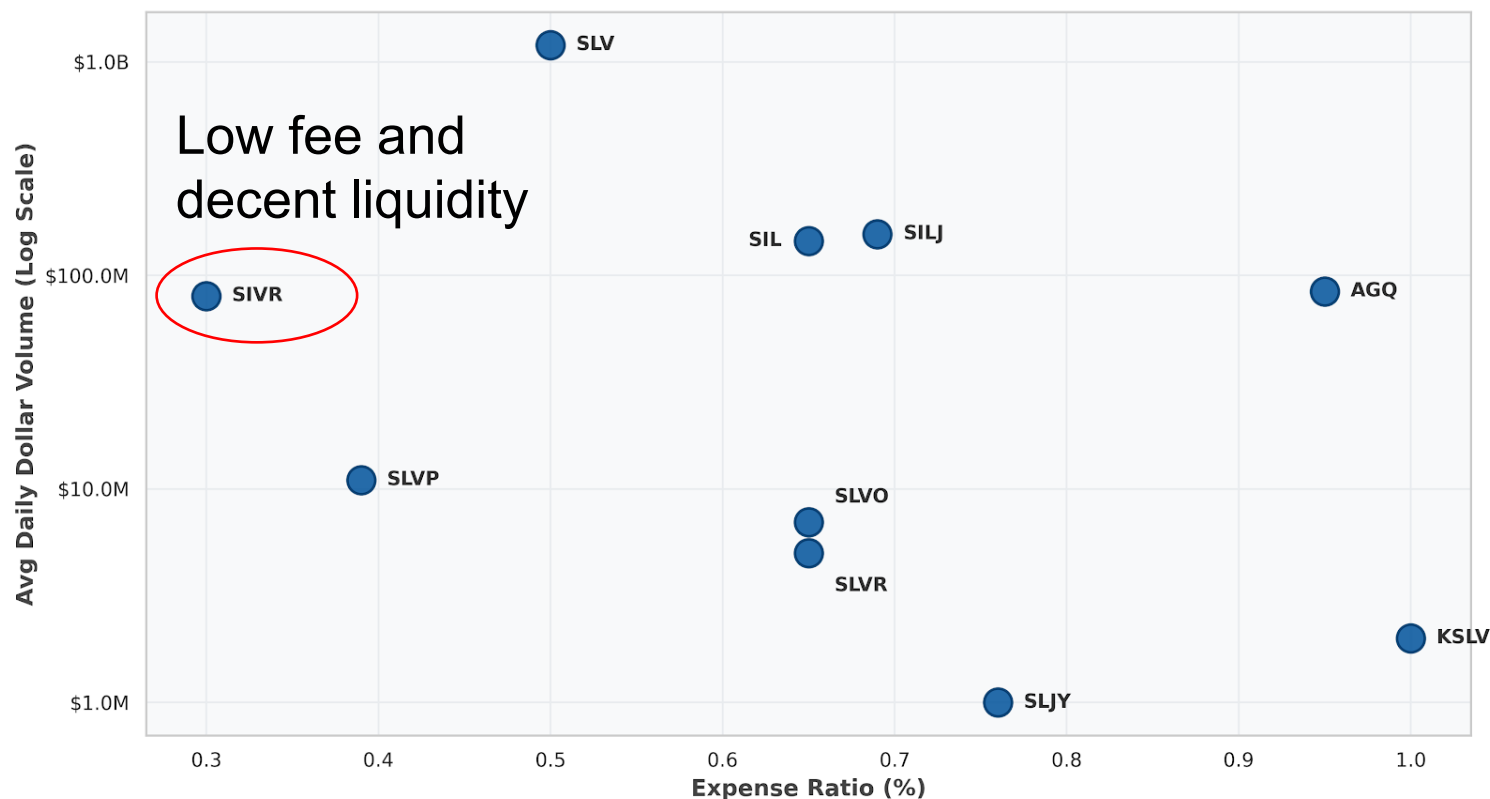
[3] <https://finance.yahoo.com/quote/GC%3DF/>; <https://finance.yahoo.com/quote/SI%3DF/>

Disclaimer: Historical data is for reference only and does not indicate future performance

Method of Silver Investing – ETF: SIVR

We chose ETFs for silver because it gives us an easier way to invest rather than storing physical silver and having to deal with futures.

Silver ETFs: Expense Ratio vs. Avg Daily Dollar Volume (6-Mo Est.)



Source

[1] <https://learn.apmex.com/investing-guide/silver-etfs-vs-physical-silver-why-tangible-silver-may-be-the-better-investment/>

[2] <https://www.ishares.com/us/products/239855/SLV>

[3] <https://www.aberdeeninvestments.com/en-us/investor/investment-solutions/commodities/sivr>

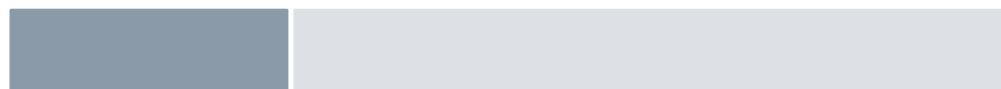
Why Silver, Not Silver Miners?

Indirect Exposure

- ✓ Mining/operating risk
- ✓ Labor & energy costs
- ✓ Management/execution risk
- ✓ Exposure to other metals

28%

of 2026 silver mine supply is expected to come from primarily silver mines



■ Primarily silver mines
■ By-product of other metals

Most silver is produced alongside other metals, so miners are an imperfect way to isolate silver exposure.

Sources:

[1] iShares Silver Trust (SLV), Silver Institute, 2026 Silver Outlook

[2] Durrett, D. (2010). How to Invest in Gold and Silver: A Complete Guide with a Focus on Mining Stocks.

Silver's Higher Reward Comes With More Risk

➤ Industrial demand

About 650M oz of 2026 demand is industrial — a recession hits silver in a way it does not hit gold.

➤ Substitution

Solar manufacturers keep cutting silver per panel, PV silver demand is expected to decline, and technology keeps reducing silver per unit.

➤ Macro pressure

Higher real yields and a stronger U.S. dollar weigh on the price; lower real yields support precious metals.

Higher Volatility

3-year annualised volatility

Silver (SLV)



Gold (GLD)



Why include gold-mining equities at all? – Performance

20-Year Cumulative Performance Comparison (Base = 100) (2016-08-29 to 2026-08-28)



“Most of the price increase in gold or silver goes directly to profit. And these higher profit margins can make stock prices spike.”

– Don Durrett

Source:

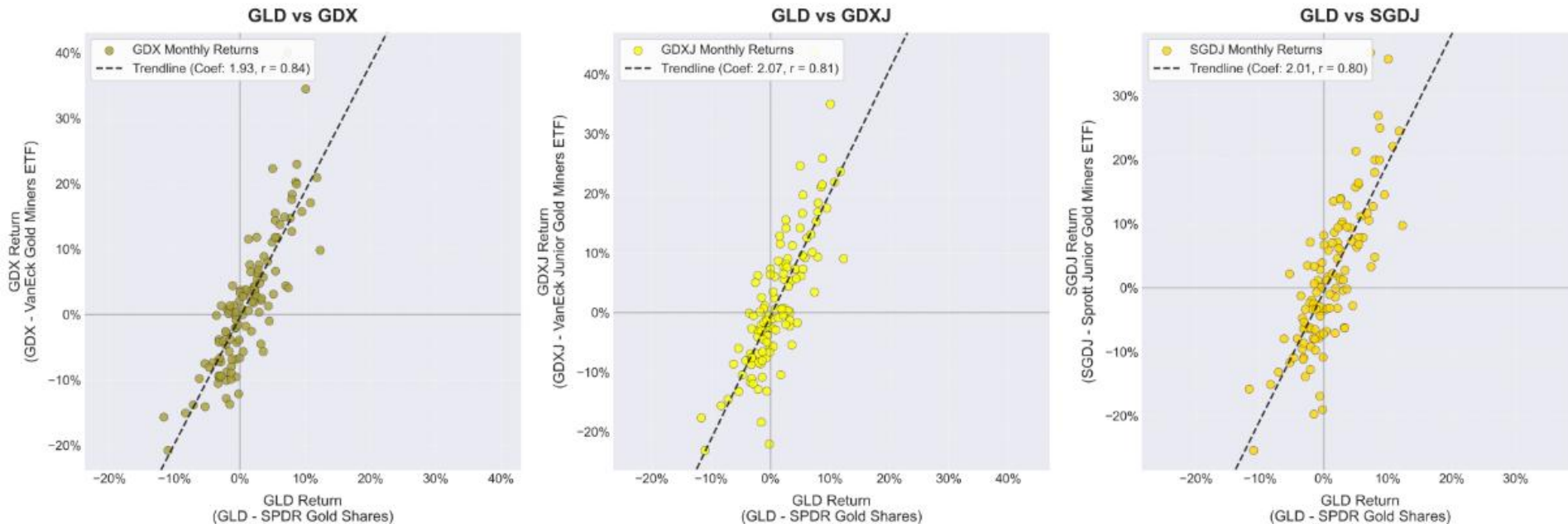
[1] VanEck (fund data); Durrett, D. (2023), How to Invest in Gold & Silver, 9th ed., ch. 6.

[2] GitHub: <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Historical%20performance%20-%20GLD%20GDX%20GDXJ%20SGDJ.ipynb>

Disclaimer: Historical data is for reference only and does not indicate future performance

Why include gold-mining equities at all? – Risk & Reward

Monthly Return Correlation: Physical Gold vs. Miners (Aug 2016 - Aug 2026)



This involves operating leverage:
Gold price +20% $\Rightarrow$ Margin per ounce +52%

Source:

- [1] Illustrative scenario based on Durrett, D. (2023), How to Invest in Gold & Silver, 9th ed., ch. 6.
- [2] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Monthly%20return%20correlation%20-%20GLD%20GDX%20GDXJ%20SGDJ.ipynb>

Explaining the return of GDX – Two-Factor OLS Regression with GLD and VT

(From 2006-08-29 to 2026-08-28)

$$\underbrace{R_{GDX,t} - R_{f,t}}_{\text{Return of GDX}} = \alpha + \underbrace{\beta_1(R_{GLD,t} - R_{f,t})}_{\substack{\text{Return of GLD} \\ \text{Beta} = 1.71}} + \underbrace{\beta_2(R_{VT,t} - R_{f,t})}_{\substack{\text{Return of VT} \\ \text{Beta} = 0.50}} + \epsilon_t$$

$R^2 = 0.644$: 64.4% of GDX's daily excessive return are explained by gold and overall stock market

GDX acts as a high-beta derivative of GLD.

Source: <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/GDX%20Return%20regression.ipynb>

Gold Mining: Comparing Producer Tiers

How cost structure, reserves, growth, and balance-sheet strength shift across the sector

Large

ALL-IN SUSTAINING COST



LOW

~\$1,413/oz



RESERVE LIFE



10–15 YRS

Long-dated reserves



PRODUCTION GROWTH



LOW TO NEGATIVE

Mature production base



BALANCE SHEET



STRONG

Net cash / low debt



Mid-Tiers & Juniors

ALL-IN SUSTAINING COST



MODERATE

\$1,413–\$1,709/oz



RESERVE LIFE



6–10 YRS

Mid-dated reserves



PRODUCTION GROWTH



STRONG

Scaling production



BALANCE SHEET



MIXED

Project debt & streaming



Small Producers,

ALL-IN SUSTAINING COST



HIGH

+\$1,700/oz



RESERVE LIFE



RESOURCE STAGE

Pre-production



PRODUCTION GROWTH



OPTIONALITY

Exploration upside



BALANCE SHEET



WEAKEST

<18-mo liquidity, many



Source: Investment Committee Memorandum: Gold-Mining Equity Allocation – Senior, Mid-Tier and Junior Miners: GDX, GDXJ and SGDJ.

Gold-miner ETF liquidity: GDX dominates trading capacity

GDX Senior miners
c.25.9m shares/day



GDXJ Junior and mid-tier miners
c.5.45m shares/day



SGDJ Small/junior miners
c.46k shares/day



Expected share volume declines sharply among the three ETFs, with SGDJ offering materially less secondary-market liquidity.

Source: Bloomberg L.P., Liquidity Trends—Expected Daily Volume, trailing nine months; accessed 28 August 2026. Metric shown is expected daily share volume

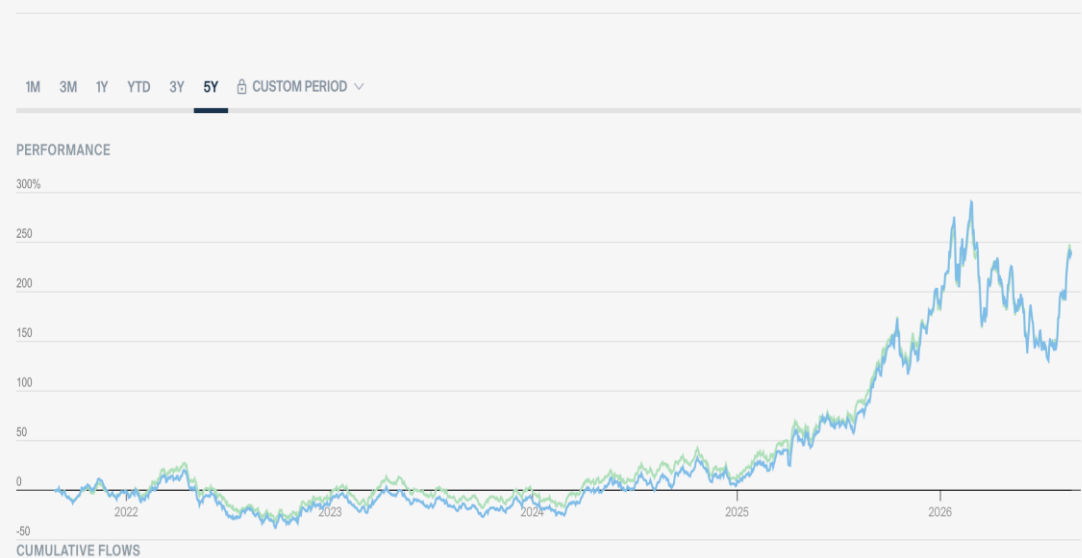
GDX (Large miners) vs GDXJ (Mid & Junior miners)

Historic performance and flows

Data as of 08/27/2026

		1M	3M	YTD	1Y	3Y	5Y
Perf.	GDX	+36.89%	+20.88%	+20.49%	+69.97%	+270.71%	+241.00%
	GDXJ	+36.33%	+19.01%	+17.94%	+77.65%	+301.00%	+238.92%
Flows	GDX	+\$375M	+\$404M	+\$1.03B	+\$1.38B	-\$3.30B	-\$3.79B
	GDXJ	+\$408M	+\$302M	-\$814M	-\$828M	-\$2.37B	-\$2.43B

GDX vs GDXJ performance and flow charts



- 1Y-3Y: GDXJ has outperformed GDX.
- 5Y: GDX 241.00%, vs GDXJ 238.92%
- **Key takeaway:** GDXJ demonstrated higher returns over the 1Y and 3Y period, while the 5Y results are nearly equal.

Source: ETF Central, "ETF Comparison: VanEck Gold Miners ETF (GDX) versus VanEck Junior Gold Miners ETF (GDXJ)"; accessed 28 August 2026.

Volatility and Drawdown (GDX vs GDXJ)

Volatility and drawdown

		3M	1Y	3Y	5Y
Volatility	GDX	+50.42%	+48.03%	+37.18%	+35.21%
	GDXJ	+53.69%	+51.10%	+39.72%	+37.99%
Max drawdown	GDX	-20.72%	-38.82%	-38.82%	-46.13%
	GDXJ	-22.56%	-40.95%	-40.95%	-48.23%
Max drawdown duration	GDX	67d	179d	179d	855d
	GDXJ	68d	179d	179d	890d

Volatility:

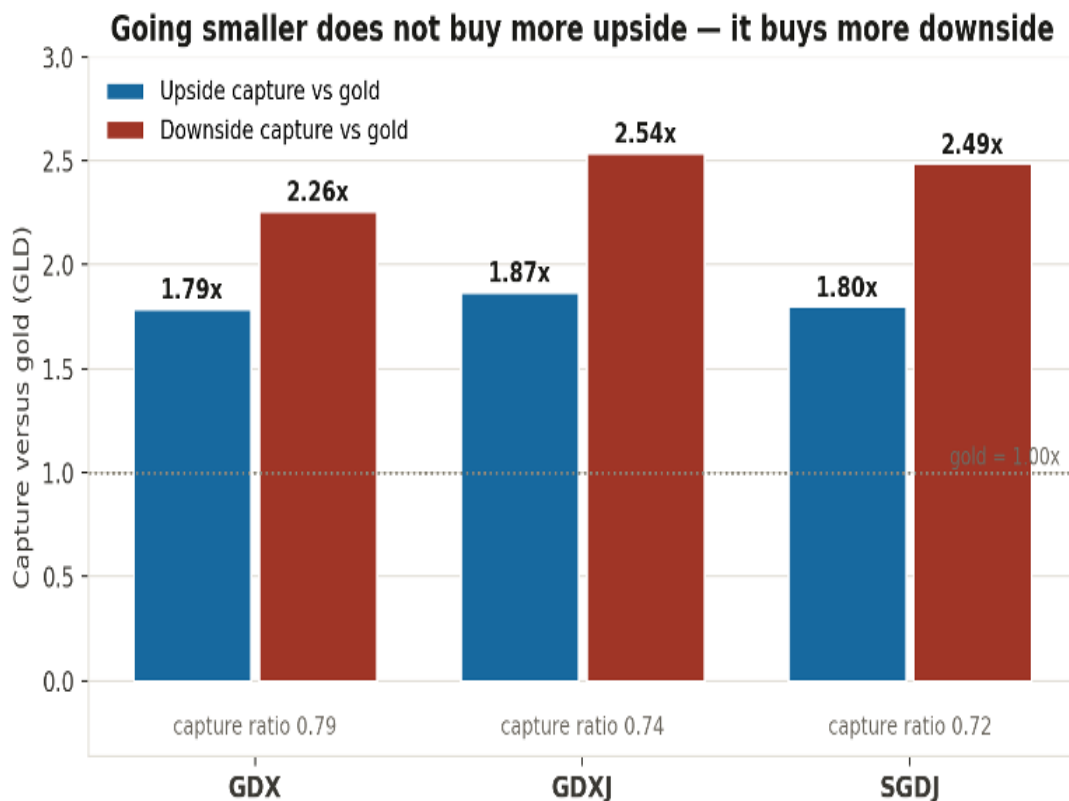
- Both are volatile.
- However, GDX has indicated lower volatility than GDXJ over all periods, indicating stable price fluctuations.

Drawdown:

- Over a five-year period, GDX lost a maximum of 46.13%, while GDXJ fell 48.23%.
- Therefore, GDX had limited downside risk.

Source: ETF Central, "ETF Comparison: VanEck Gold Miners ETF (GDX) versus VanEck Junior Gold Miners ETF (GDXJ)"; accessed 28 August 2026.

Why we exclude small-cap gold miners (SGDJ)



What the evidence shows

- ⇒ **No Additional Upside 1.80x**
SGDJ upside capture is effectively the same as GDX's 1.79x
- ⇒ **Greater Downside 2.49x**
Smaller miners participate much more heavily when gold declines
- ⇒ **Worst Capture Ratio 0.72x**
SGDJ: 0.72 upside per downside—the weakest trade-off.

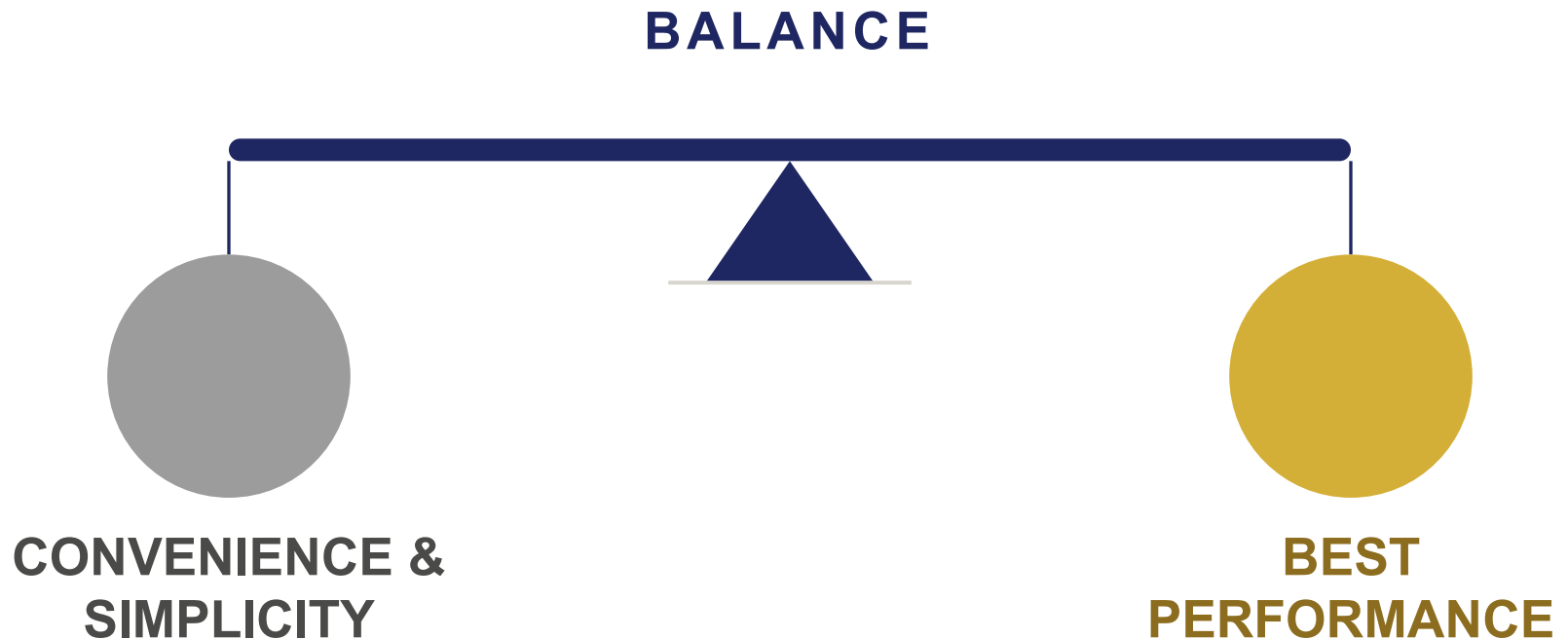
Source:

[1] Monthly total returns for GLD, GDX, GDXJ and SGDJ, StockAnalysis <https://stockanalysis.com/etf/>

[2] Validated against Total Real Returns <https://totalrealreturns.com/>

Portfolio management strategy

Adjusted Monthly — Built for Shifting Market Cycles



Source: DeMiguel, V., Garlappi, L., & Uppal, R. (2007). Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy?. *The Review of Financial Studies*, 20(6), 1915-1953.

Portfolio strategies besides monthly adjustment?

A signal-driven approach to precious metal allocation

Silver Fundamentals / Gold Silver Ratio

- Industrial Demand, Supply/market deficit

Market Volatility

- VIX/liquidity conditions

Macro Environment

- 10-year real yields, U.S. dollar

Scenario analysis

- Learn from the past and get prepared

All great investments are **situational**

Portfolio adjustment watch 1 – Based on Gold Silver Ratio

Historical Performance: Gold, Silver, and Gold/Silver Ratio (Jan 2001 – Aug 2026)



OW: Overweight

Source:

[1] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Gold%20Silver%20Ratio.ipynb>

[2] <https://finance.yahoo.com/quote/GC%3DF/>; <https://finance.yahoo.com/quote/SI%3DF/>

Disclaimer: Historical data is for reference only and does not indicate future performance

Portfolio adjustment watch 2 - VIX

Historical Performance: GLD, Silver, and GDX relative to VIX
(Aug 29, 2006 – Aug 28, 2026)



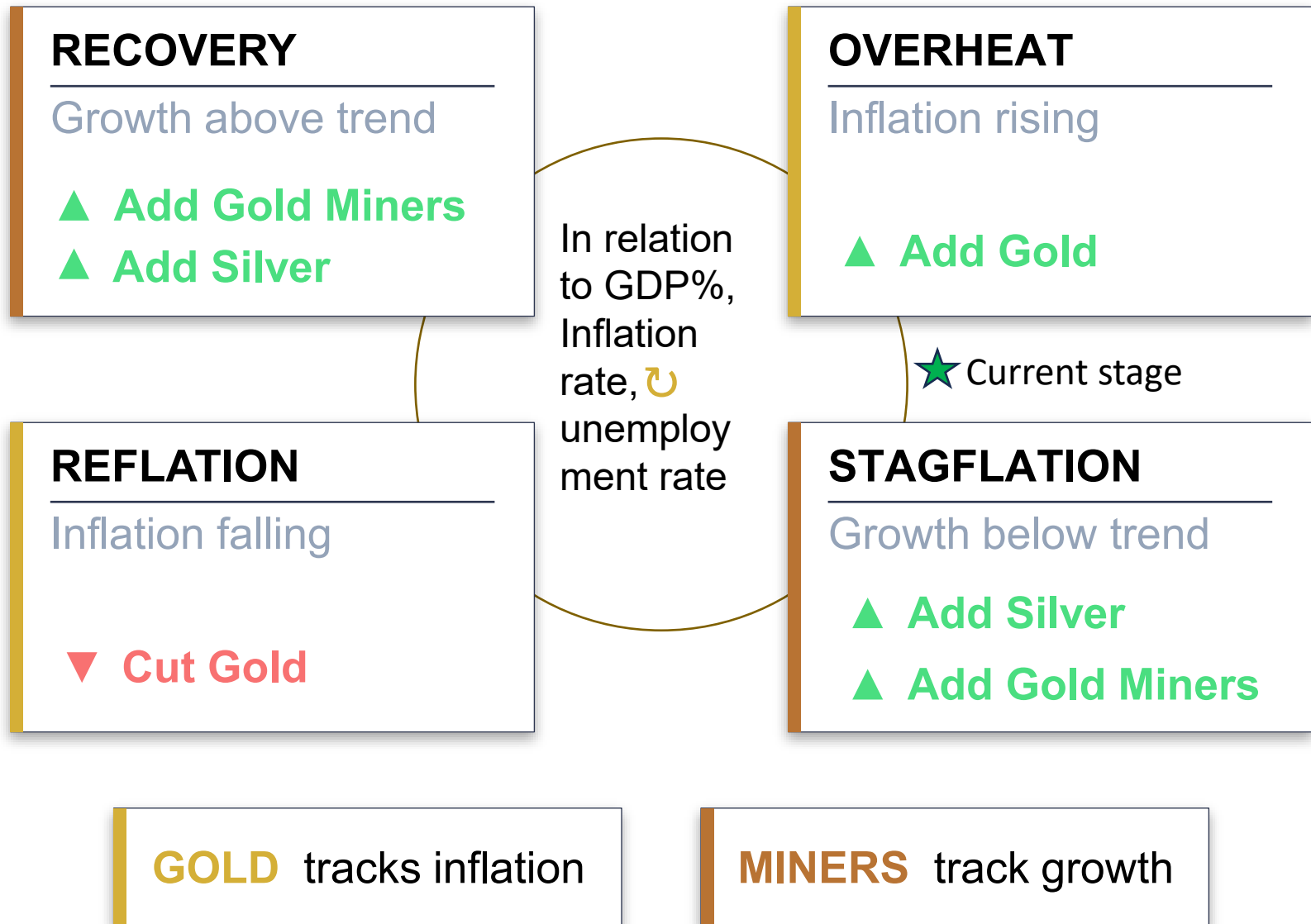
OW: Overweight

Source:

[1] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Portfolio%20VIX.ipynb>

Disclaimer: Historical data is for reference only and does not indicate future performance

Portfolio adjustment watch 3 - Across the Macro Cycle



Scenario analysis 1: Falling real yield / weak dollar (Gold)

Gold vs US real yields (reversed), in yoy changes



Projection: As real yields fall, we expect gold prices to hold stable growth.

If gold prices fall below support level, then we need to exit or reduce exposure.

- 2009-2012: real yields fell due to QE and monetary policies; gold prices nearly doubled during this time.
- 2020-2021: real yields fell during COVID-19, while gold reached an all-time high.

Source: J. Safra Sarasin. *All That Glitters Is Not Gold*. Cross-Asset Weekly. 28 February 2025. Accessed 29 August 2026.

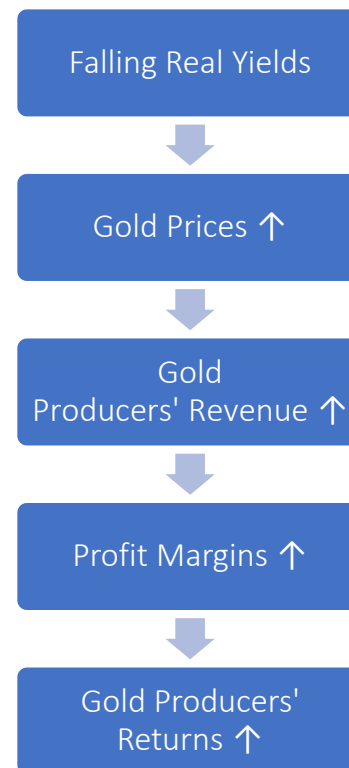
Jury, Stephen. *Is it a Golden Era for Gold?* J.P. Morgan Private Bank. Published 9 February 2026. Accessed 29 August 2026.

Scenario analysis 1: Falling real yield / weak dollar (Gold Mining)

- Higher gold prices increase gold producers' revenues, while operating costs remain relatively fixed.

Example 1: Newmount Corporation (2020)

- Revenue increased 18%
 - Avg realised gold price increased from US\$1399/oz to \$1775/oz.
 - Adjusted EBITDA increased 48%.
- Projection:** As real yields fall and gold prices hold stable growth, we expect gold mining equities to follow, benefiting from higher profit margins and earnings.



Source: Hidayat, Muflih. *Gold Producers Operating Leverage: How It Works in 2026*. Discovery Alert, 27 May 2026. Accessed 29 August 2026.

Newmount Corporation. *Newmount Delivers Record Full-Year and Fourth Quarter Results*. 19 February 2021. Accessed 29 August 2026.

Scenario analysis 1: Falling real yield / weak dollar (Silver)

Silver vs US real yields (reversed), in yoy changes



Projection: As real yields fall, we expect silver prices to continue a stable growth, supported by both investment and industrial demand.

- 2010-2011: real yields fell following QE after the financial crisis; silver surged.
- 2023-2025: real yields fell from their peak, silver surged higher.

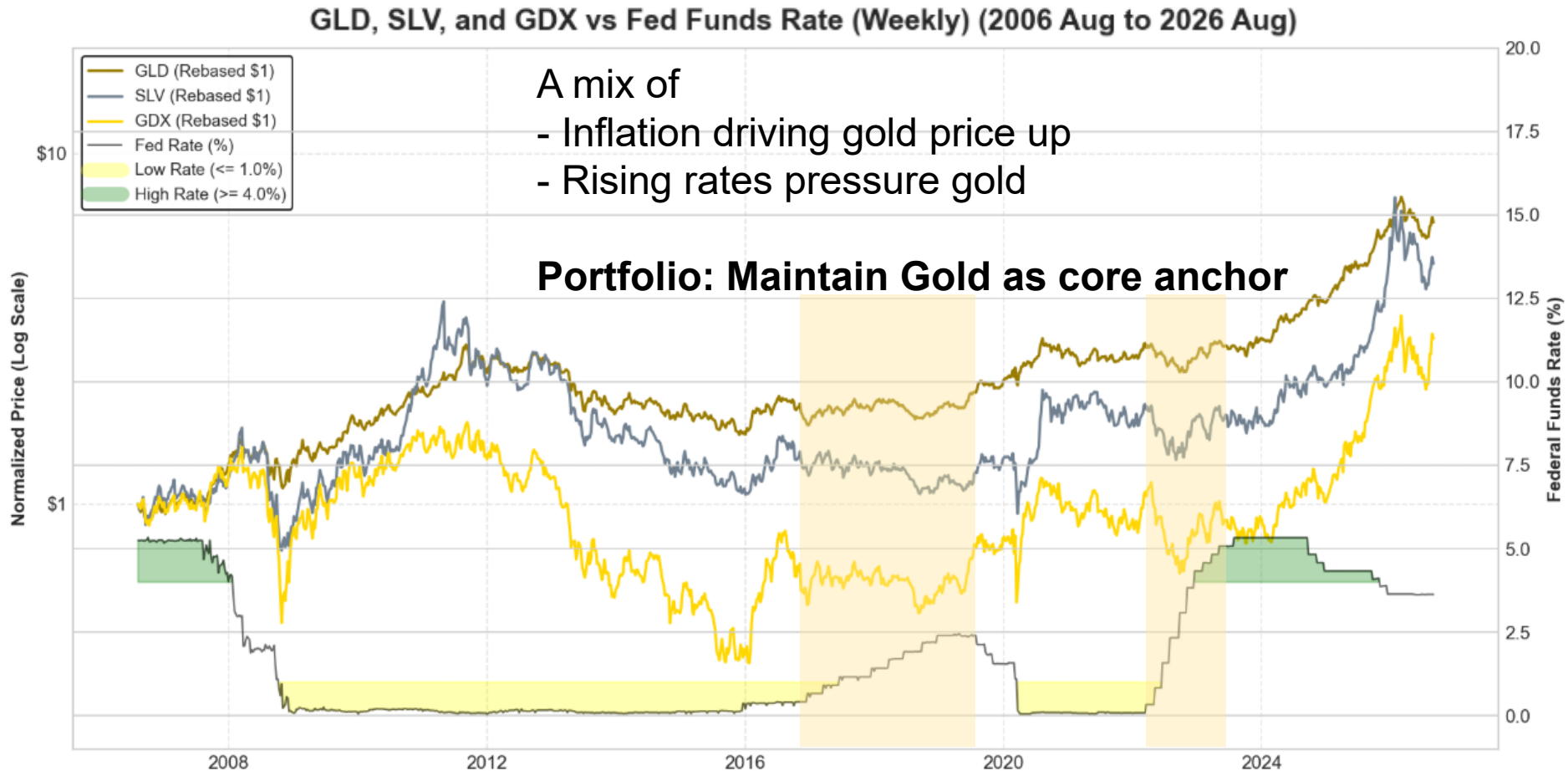
RECOMMENDED ALLOCATION

50% Gold / 20% Silver / 30% Gold Mining

Source: J. Safra Sarasin. *All That Glitters Is Not Gold*. Cross-Asset Weekly. 28 February 2025. Accessed 29 August 2026.

Tomlinson, Mitchell. Key Events, Price Drivers and Silver's Performance Over the Last 20 Years. Gold Bullion Australia. 27 December 2024. Accessed 29 August 2026.

Scenario analysis 2: Inflation with Hawkish Monetary Policy

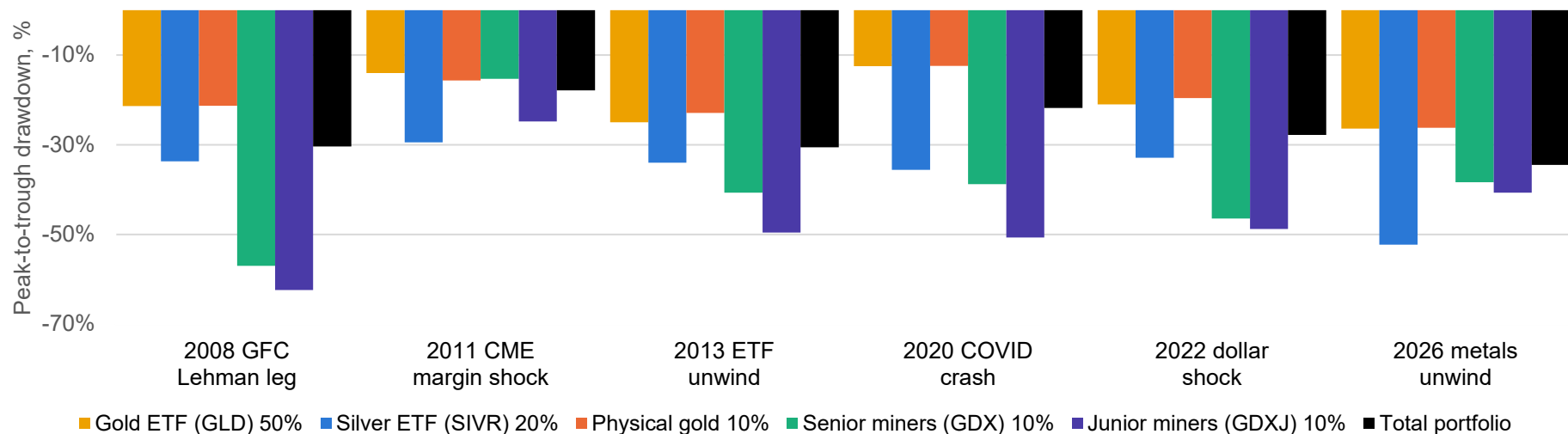


Source:

[1] <https://fred.stlouisfed.org/series/DFF>

[2] <https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/blob/main/Portfolio%20under%20Fed%20rate.ipynb>

Scenario analysis 3: Liquidity shocks: evidence from six historical episodes



Sleeve · weight	2008 GFC Lehman leg	2011 CME margin shock	2013 ETF unwind	2020 COVID crash	2022 Dollar shock	2026 Metals unwind	Average
Gold ETF (GLD) 50%	-21.4	-14.0	-25.0	-12.5	-21.0	-26.4	-20.1
Silver ETF (SIVR) 20%	-33.7	-29.5	-34.0	-35.6	-32.9	-52.3	-36.3
Physical gold 10%	-21.3	-15.7	-22.9	-12.4	-19.6	-26.2	-19.7
Senior miners (GDX) 10%	-57.0	-15.3	-40.7	-38.8	-46.5	-38.4	-39.5
Junior miners (GDXJ) 10%	-62.4	-24.8	-49.6	-50.7	-48.8	-40.7	-46.2
TOTAL PORTFOLIO · 100%	-30.4	-17.9	-30.6	-21.8	-27.8	-34.5	-27.1

S&P 500 over the same window
(memo)

-33.0 -7.1 -5.6 -33.7 -20.6 -8.9 -18.1

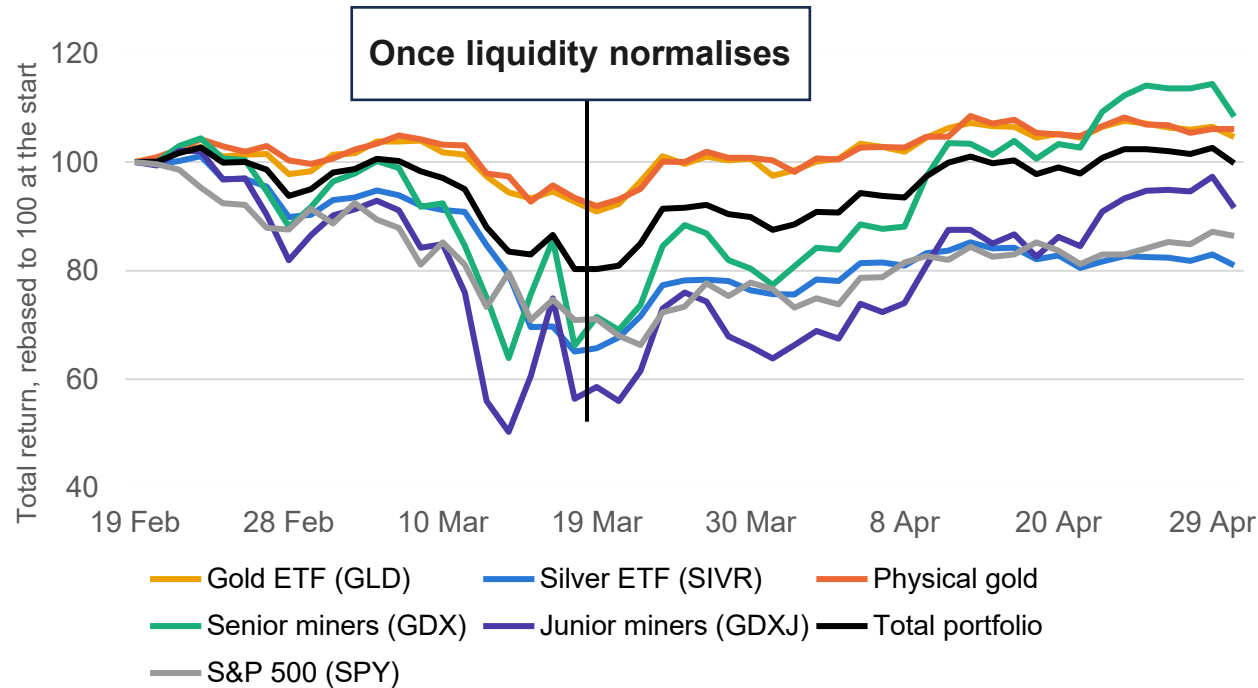
Source:

[1] <https://www.lbma.org.uk/prices-and-data/lbma-precious-metal-prices>

[2] <https://finance.yahoo.com/quote/GLD/>

Disclaimer: Historical data is for reference only and does not indicate future performance

COVID-19 forced selling: gold miners recovered faster after the panic



KEY FINDINGS

-21.8%

Portfolio drawdown

-12.5%

Gold drawdown

-36% to -51%

Silver and miner losses

Fastest recovery

Gold returned above 100 first

COVID dash for cash, 19 Feb – 30 Apr 2020. Gold troughs -12.5% on 19 March and is back above its pre-crisis level by 9 April; silver, seniors and juniors fall 36%, 39% and 51%.

Recommended allocation

During the unwind 55% GLD / 10% physical / 10% SIVR / 15% GDX / 5% GDXJ / 5% cash

Once liquidity normalizes 45% GLD / 10% physical / 15% SIVR / 15% GDX / 15% GDXJ / 0% cash

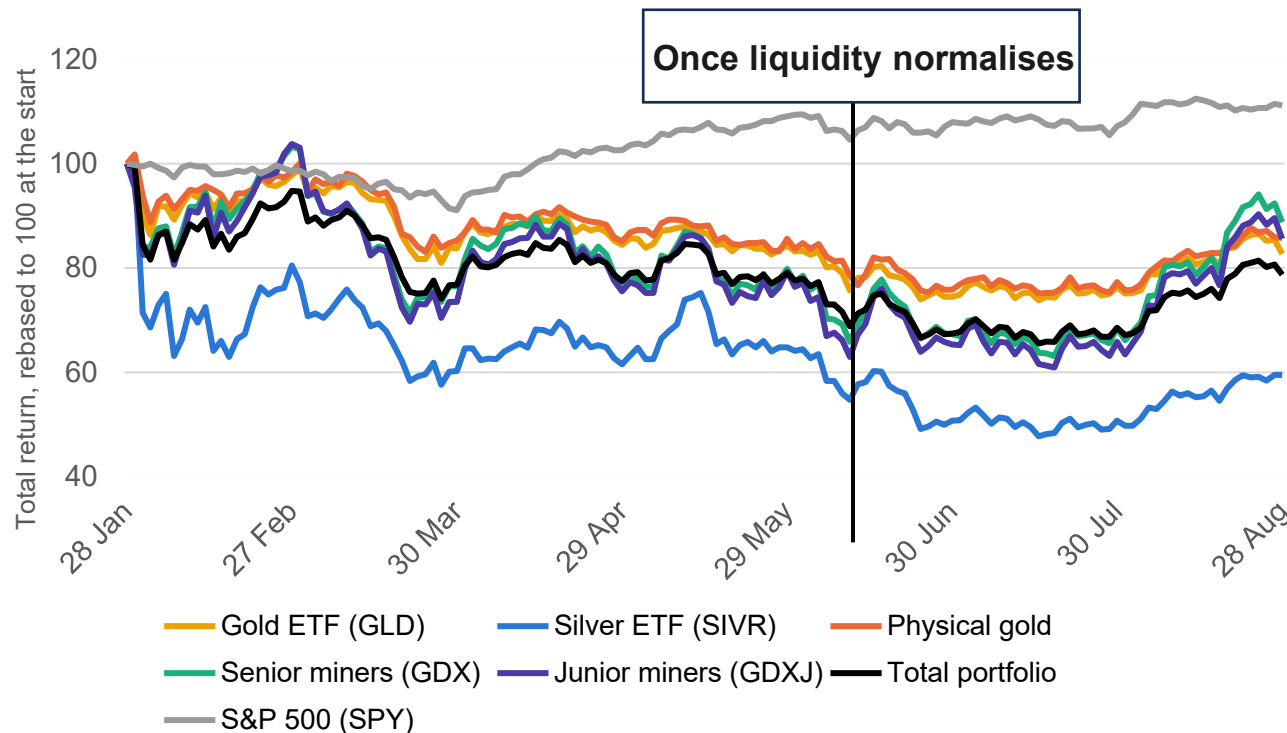
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[1] <https://www.lbma.org.uk/prices-and-data/lbma-precious-metal-prices>

[2] <https://finance.yahoo.com/quote/GLD/>

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Liquidity of metals — 2026 metal unwind review



KEY FINDINGS

-34.5%

Portfolio drawdown

-15.1%

Worst single day

-52.3%

Silver sleeve loss

Still unrecovered

**21.3% below the
January peak**

2026 metals unwind, 28 Jan – 16 Jul 2026 (chart to 28 Aug). The portfolio fell 34.5% while the S&P 500 rose; silver -52.3%, juniors -40.7%, seniors -38.4%.

Recommended allocation

During the unwind 55% GLD / 10% physical / 10% SIVR / 15% GDX / 5% GDXJ / 5% cash

Once liquidity normalizes 45% GLD / 10% physical / 15% SIVR / 20% GDX / 5% GDXJ / 5% cash

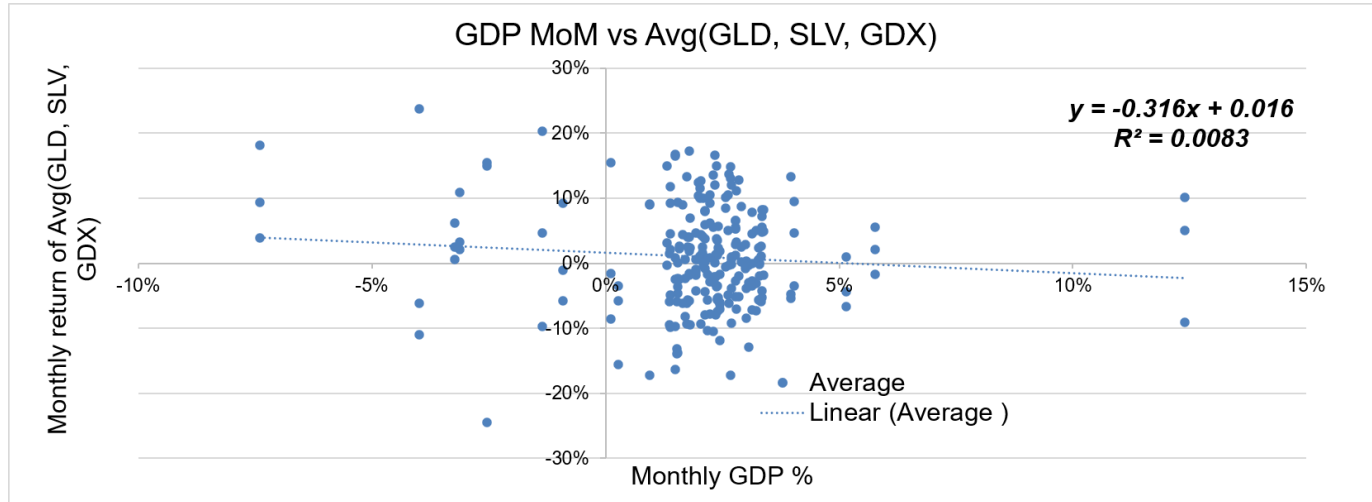
Source:

[1] <https://www.lbma.org.uk/prices-and-data/lbma-precious-metal-prices>

[2] <https://finance.yahoo.com/quote/GLD/>

Disclaimer: Historical data is for reference only and does not indicate future performance

Scenario analysis 4: Performance under strong economy



GDP MoM% is not a strong indicator of our return.
(Low R^2)

However, metals act as a hedge under different GDP scenarios especially under weak ones supporting our portfolio as long term investment across cycles.

From 2006 Oct to 2026 Jun	GDP Monthly Return %	GLD Avg Monthly	SLV Avg Monthly	GDX Avg Monthly
Strong (MoM >2.5%)	3.54%	0.96%	1.48%	0.98%
Normal (MoM 0~2.5%)	1.84%	0.70%	0.11%	0.15%
Weak (MoM <0%)	-3.22%	1.83%	5.47%	5.10%

Overweight SLV & GDX under extremes (Overheat & Recession)
Maintain GLD as majority under normal circumstances

Source:

[1] Yahoo Finance: <https://finance.yahoo.com/quote/GLD>;
<https://finance.yahoo.com/quote/SLV>; <https://finance.yahoo.com/quote/GDX>

[2] <https://tradingeconomics.com/united-states/gdp-growth>

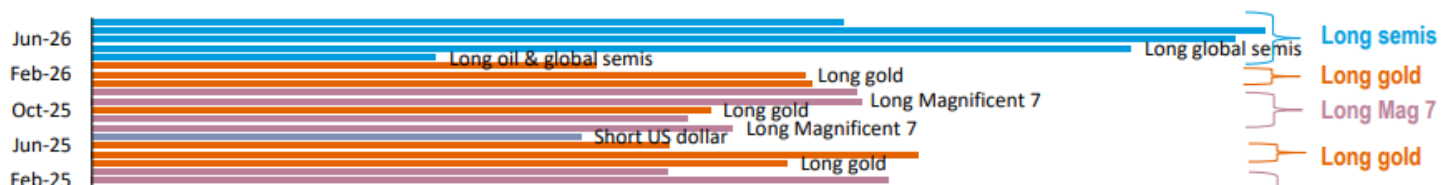
Scenario analysis 5: When long gold become overcrowded - Crisis of Gold mining companies – 2026 sell off

20-Year Cumulative Performance Comparison (Base = 100) (2006-08-29 to 2026-08-28)



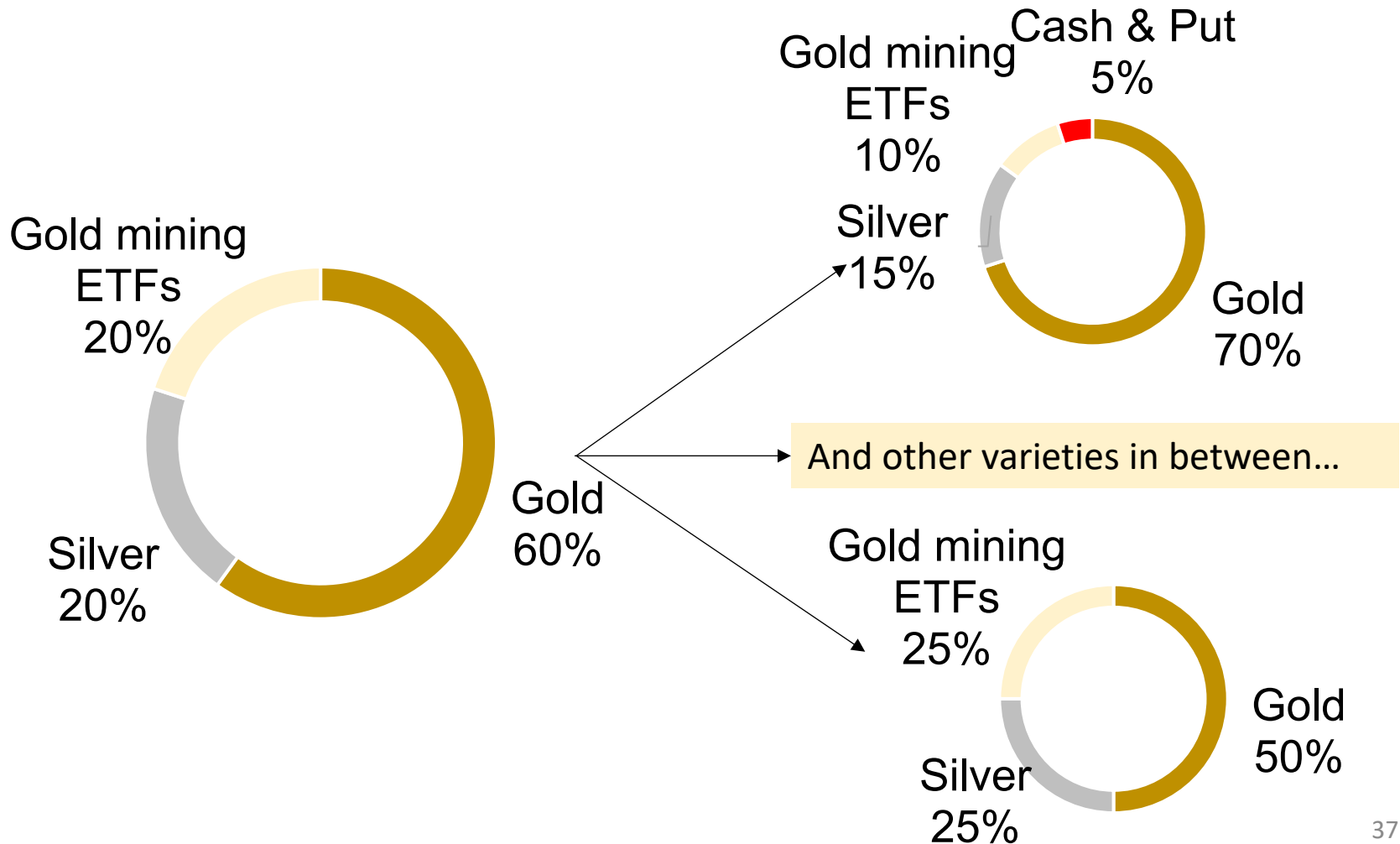
Chart 24: Evolution of Global FMS "most crowded trade"

History of Global FMS "most crowded trade" answers



Conclusion:

The 60-20-20 allocation is based on what we think right now in the long term. More importantly is to adapt to the market trend and adjust accordingly.



Appendix 1 – Prompts used for generating the visualization code & GitHub folder for codes and visualization

Price Visualization:

https://gemini.google.com/share/d/1uC9twNNOIg7RQV4U8cb4i1G_NTV81P6?usp=sharing

<https://gemini.google.com/share/d/1p-sdLj-JvkAWluuDXU19WNJDjigHeHeK?usp=sharing>

<https://gemini.google.com/share/d/1U9jvmHSAi5k3yfQ0U5Relzl32M8NLw0?usp=sharing>

Correlation calculation:

https://gemini.google.com/share/d/1vpt1x0dsY_REAomC_8BLO4RoTgKZMMCQ?usp=sharing

GDX regression:

<https://gemini.google.com/share/d/15Z46EhsuJKldpJmipjQYcBjMkeXU4GWD?usp=sharing>

Portfolio Fed:

https://gemini.google.com/share/d/1DkvCby0duGRI3zR_fmP3BawhGdHTJKqf?usp=sharing

GDP data:

<https://www.perplexity.ai/search/826ecc84-fca7-448b-a795-0772eff624ff>

GitHub folder:

<https://github.com/Xiaowen-CHEN-Learner/FMR-A---Structuring-a-1-Million-Portfolio-Gold-Silver-and-Gold-Mining-Equities/tree/main>

Disclaimer: The code generated using AI was for reference only, there were manual review and adjustment for the code used in the project thus having differences. Please refer to GitHub link in appendix 2 for the final code for the visualization.

Appendix 2 – GDX Two-Factor OLS Regression Result

--- Two-Factor OLS Regression Results ---

OLS Regression Results

```
=====
Dep. Variable:          GDX_excess    R-squared:                0.644
Model:                  OLS           Adj. R-squared:          0.644
Method:                 Least Squares  F-statistic:             4128.
Date:                  Sun, 30 Aug 2026  Prob (F-statistic):       0.00
Time:                  13:21:13        Log-Likelihood:          12474.
No. Observations:      4568           AIC:                    -2.494e+04
Df Residuals:          4565           BIC:                    -2.492e+04
Df Model:               2
Covariance Type:       nonrobust
=====
```

	coef	std err	t	P> t	[0.025	0.975]
const	-0.0005	0.000	-2.023	0.043	-0.001	-1.47e-05
GLD_excess	1.7058	0.021	82.264	0.000	1.665	1.746
VT_excess	0.4985	0.018	27.395	0.000	0.463	0.534

```
=====
Omnibus:                1391.883    Durbin-Watson:           2.275
Prob(Omnibus):           0.000      Jarque-Bera (JB):        96654.959
Skew:                    0.563      Prob(JB):                0.00
Kurtosis:                25.507     Cond. No.                91.0
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

--- Model Effectiveness & Significance Summary ---

```
R-squared (Explanatory Power): 0.6439 (64.39%)
Gold Factor (GLD) Beta:       1.7058 (p-value: 0.0000e+00)
Market Factor (VT) Beta:      0.4985 (p-value: 4.1944e-153)
```